CHP-9 The Magnetic Force and Field

General Physics 2 2026-11-02

The magnetic force and field

The magnetic force was known for many centuries, but only in the 1800s did experimenters discover that electric currents also exert magnetic forces on magnets and current-carrying wires.

Between two charges in motion, there acts not only the Coulomb force but also a force that depends on their velocities.

Our development differs from the historical development: we first discuss the magnetic force between two charges, then the force between current-carrying wires.

The magnetic force

Charges at rest and charges in motion

According to Coulomb's law, the electric force between charges q and q' is

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{r^2}\hat{r}$$

This is the only force when the charges are at rest. When they are moving, there is an extra force: the magnetic force.

Its magnitude and direction depend on the distance and on the velocities of the particles. In general, this dependence is fairly complicated.

Two charges moving parallel to each other

To simplify the discussion, consider charges q and q' moving parallel to each other with velocities $\vec{v}$ and $\vec{v}'$.

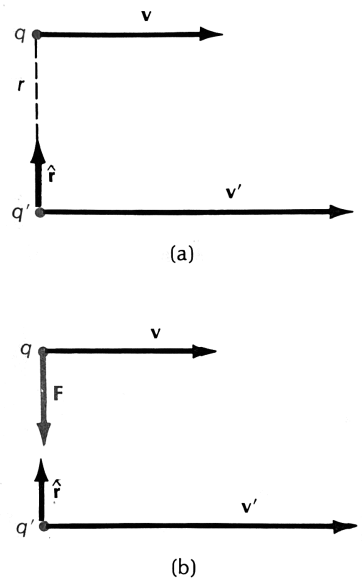


Fig. 30.1 (a) Two point charges q and q' instantaneously moving side by side, with parallel velocities at right angles to $\hat{\mathbf{r}}$. (b) The magnetic force on q is attractive.

The magnetic force on q exerted by q' is

$$\vec{F} = -\,\text{constant}\,\frac{qq'}{r^2}vv'\hat{r}$$

For charges of equal sign, the force is attractive if the velocities are paralel and repulsive for antiparallel velocities.

The magnetic force is zero unless both charges are in motion. The force acts only if both charges are in motion.

The permeability constant

In SI units, the constant is conventionally written as $\frac{\mu_0}{4\pi}$.

$$\mu_0 \approx 4\pi \times 10^{-7} \frac{\mathrm{N\,s^2}}{\mathrm{C^2}} \approx 1.26 \times 10^{-6} \frac{\mathrm{N\,s^2}}{\mathrm{C^2}}$$

It is called the permeability constant.

$$\vec{F} = -\frac{\mu_0}{4\pi} \frac{qq'}{r^2} vv' \hat{r}$$

Comparing electric and magnetic forces

For this configuration, the force magnitudes are

$$F_{\text{Coulomb}} = \frac{1}{4\pi\varepsilon_0}\frac{|qq'|}{r^2}, \quad F_{\text{mag}} = \frac{\mu_0}{4\pi}\frac{|qq'|}{r^2}vv'$$

Therefore

$$\frac{F_{\text{mag}}}{F_{\text{Coulomb}}} = \mu_0\varepsilon_0 vv' = \frac{vv'}{c^2}$$

Here $\mu_0\varepsilon_0 \approx 1.11 \times 10^{-17}\,\frac{\text{s}^2}{\text{m}^2}$ and $c \approx 3 \times 10^8\,\frac{\text{m}}{\text{s}}$ is the speed of light.

Thus the magnetic force is small compared with the Coulomb force when both speeds are small compared with c.

When magnetic forces become important

Although the magnetic force is small compared with the electric force, it becomes important when electric forces cancel.

For example, consider a moving point charge near a straight current-carrying wire.

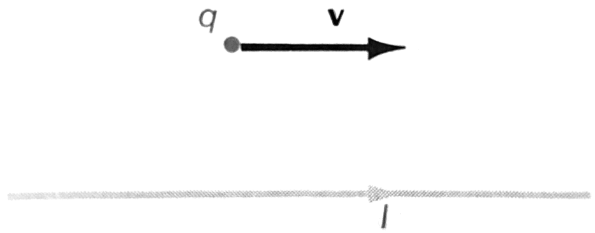
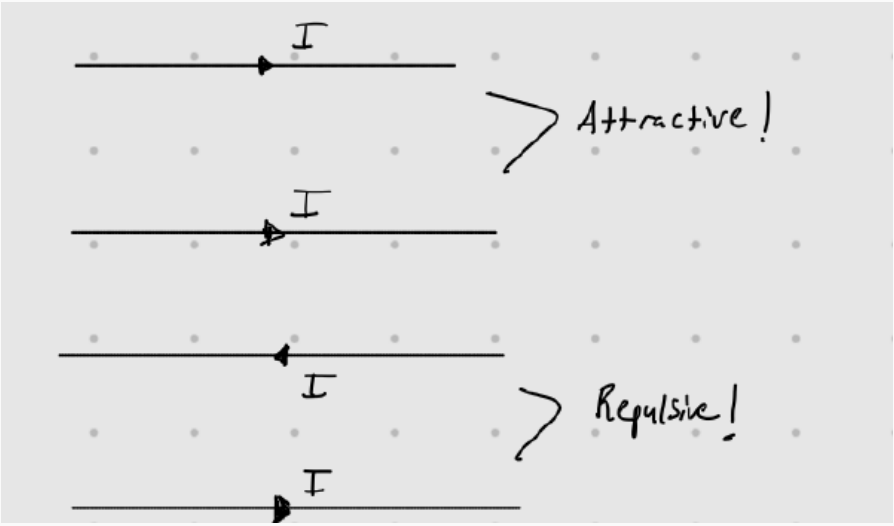


Fig. 30.2 Point charge moving parallel to a current.

The wire contains equal amounts of positive and negative charge. The number of moving electrons is very large, so the magnetic force can be large.

Forces between current-carrying wires



Parallel currents attract; antiparallel currents repel.

The magnetic force can also be regarded as arising through a transformation of reference frame applied to an electric interaction.

For two charges at rest, there is no magnetic force. In another reference frame, where the charges move, a magnetic force appears.

The general case

The velocities may make arbitrary angles with the line between the charges.

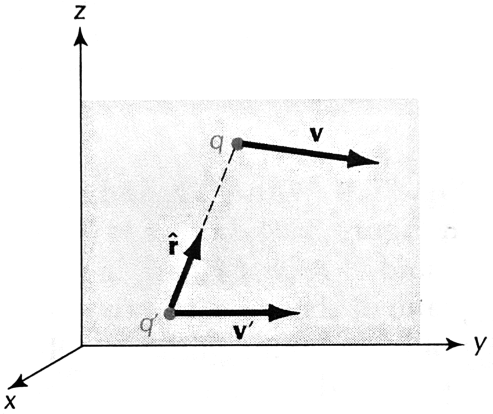


Fig. 30.4 Two point charges with velocities at different angles relative to the unit vector $\hat{\mathbf{r}}$. In this example, the point charges and the vectors $\mathbf{v}$, $\mathbf{v}'$, and $\hat{\mathbf{r}}$ are in the z–y plane, or the plane of the page.

Magnetic force on moving point charge

The magnetic force on q exerted by q' is

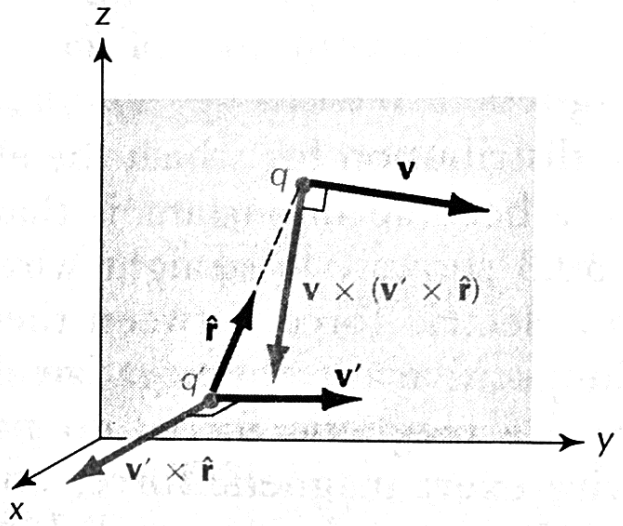
$$\vec{F} = \frac{\mu_0}{4\pi}\frac{qq'}{r^2}\vec{v} \times (\vec{v}' \times \hat{r})$$

Here $\hat{r}$ points from q' to q.

Finding the force direction

First find $\vec{v}' \times \hat{r}$ using the right-hand rule. Then take the cross product of $\vec{v}$ with that result.

Fig. 30.5 To find the direction of the cross product $\mathbf{v}' \times \hat{\mathbf{r}}$ from the right-hand rule, place the fingers of the right hand along $\mathbf{v}'$ and curl them toward $\hat{\mathbf{r}}$; the thumb then points out of the plane of the page and indicates the direction of the vector $\mathbf{v}' \times \hat{\mathbf{r}}$. Next, imagine that this vector has been shifted to the position of the charge q. Place the fingers along $\mathbf{v}$ and curl them toward the shifted vector. The thumb then indicates the direction of the cross product $\mathbf{v} \times (\mathbf{v}' \times \hat{\mathbf{r}})$, which is in the plane of the page and perpendicular to $\mathbf{v}$.



The sign of qq' determines whether to reverse the resulting direction.

The force on the other charge

Exchange the charges and velocities, with $\hat{r}' = -\hat{r}$:

$$\vec{F}' = \frac{\mu_0}{4\pi}\frac{qq'}{r^2}\vec{v}' \times (\vec{v} \times \hat{r}')$$

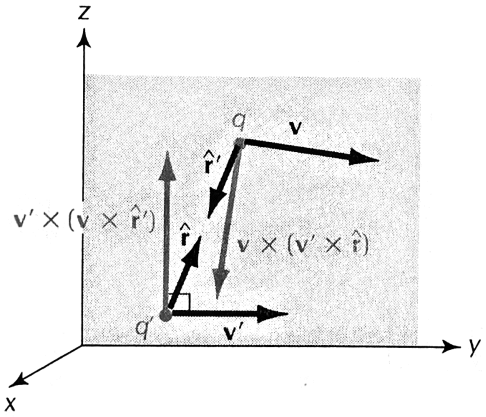


Fig. 30.6 With $\mathbf{v}'$, $\hat{\mathbf{r}}$, and $\mathbf{v}$, as in Figure 30.4, both $\mathbf{v} \times (\mathbf{v}' \times \hat{\mathbf{r}})$ and $\mathbf{v}' \times (\mathbf{v} \times \hat{\mathbf{r}}')$ are in the plane of the page, but they are not of equal magnitudes and opposite directions.

The two magnetic forces are not, in general, equal and opposite. Newton's third law does not hold for the magnetic particle forces alone.

Momentum of particles and fields

As charges move, their electric field redistributes itself: field energy flows through space. The electromagnetic field also carries momentum.

Although the magnetic particle forces do not obey Newton’s third law by themselves, the total momentum of the particles and field is conserved.

The magnetic field

Defining the magnetic field

As with the electric field, it is useful to introduce a field rather than work directly with the force between charges.

Place a test charge q at a point and give it velocity $\vec{v}$. Its magnetic force is

$$\vec{F} = q\left(\vec{v} \times \vec{B}\right)$$

This force is always perpendicular to both $\vec{B}$ and $\vec{v}$.

Units of magnetic field

The SI unit is the tesla:

$$1\ \mathrm{T} = \frac{1\ \mathrm{N}}{\mathrm{C} \cdot \frac{\mathrm{m}}{\mathrm{s}}}$$

An alternative name is weber per square metre, $\frac{\mathrm{Wb}}{\mathrm{m}^2}$.

For weak magnetic fields, a smaller unit is often used: the gauss.

$$1\ \text{gauss} = 10^{-4}\ \mathrm{T} = 10^{-4}\frac{\mathrm{Wb}}{\mathrm{m}^2}$$

The gauss is a cgs unit.

Field of a moving point charge

Comparing

$$\vec{F} = \frac{\mu_0}{4\pi}\frac{qq'}{r^2}\vec{v} \times \left(\vec{v}' \times \hat{r}\right)$$

with $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$, we find

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q'}{r^2}\left(\vec{v}' \times \hat{r}\right)$$

This is the field generated by the moving source charge q'.

Example: an electron in a hydrogen atom

In the classical orbit model, an electron moves in a circular orbit of radius $r = 5.3 \times 10^{-11}$ m at speed $v' = 2.2 \times 10^{6} \frac{\text{m}}{\text{s}}$.

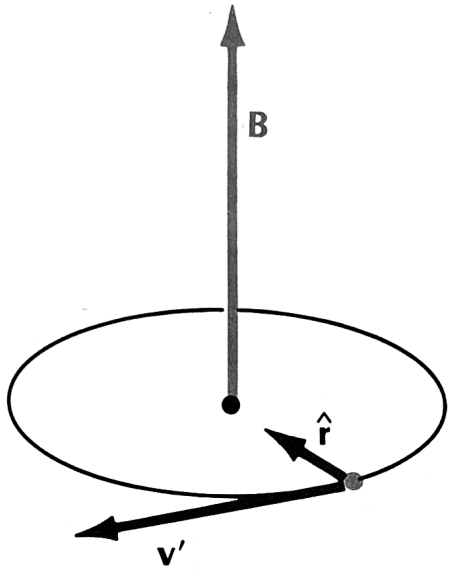


Fig. 30.7 Orbit of an electron around a nucleus, and magnetic field generated by the electron.

At the centre,

$$B = \frac{\mu_0}{4\pi}\frac{ev'}{r^2} \approx 13\ \text{T}$$

The direction shown is upward because of the electron's negative charge.

Some magnetic fields

TABLE 29.1 SOME MAGNETIC FIELDS

At surface of pulsar	$\approx 10^8$ T
Maximum achieved in laboratory:	
Explosive compression of field lines	1×10^3
Steady	45
In particle accelerator magnet	8
In large bubble-chamber magnet	2
In MRI magnet (a)	1.5
In sunspot (b)	≈ 0.3
Near small ceramic magnet	$\approx 2 \times 10^{-2}$
At surface of Sun	$\approx 10^{-2}$
Near household wiring (c)	$\approx 10^{-4}$
At surface of Earth	$\approx 5 \times 10^{-5}$
In sunlight (rms)	3×10^{-6}
In Crab Nebula (d)	$\approx 10^{-8}$
In radio wave (rms)	$\approx 10^{-9}$
In interstellar galactic space	$\approx 10^{-10}$
Produced by human body	3×10^{-10}
In shielded antimagnetic chamber	2×10^{-14}

Magnetic field lines of a moving charge

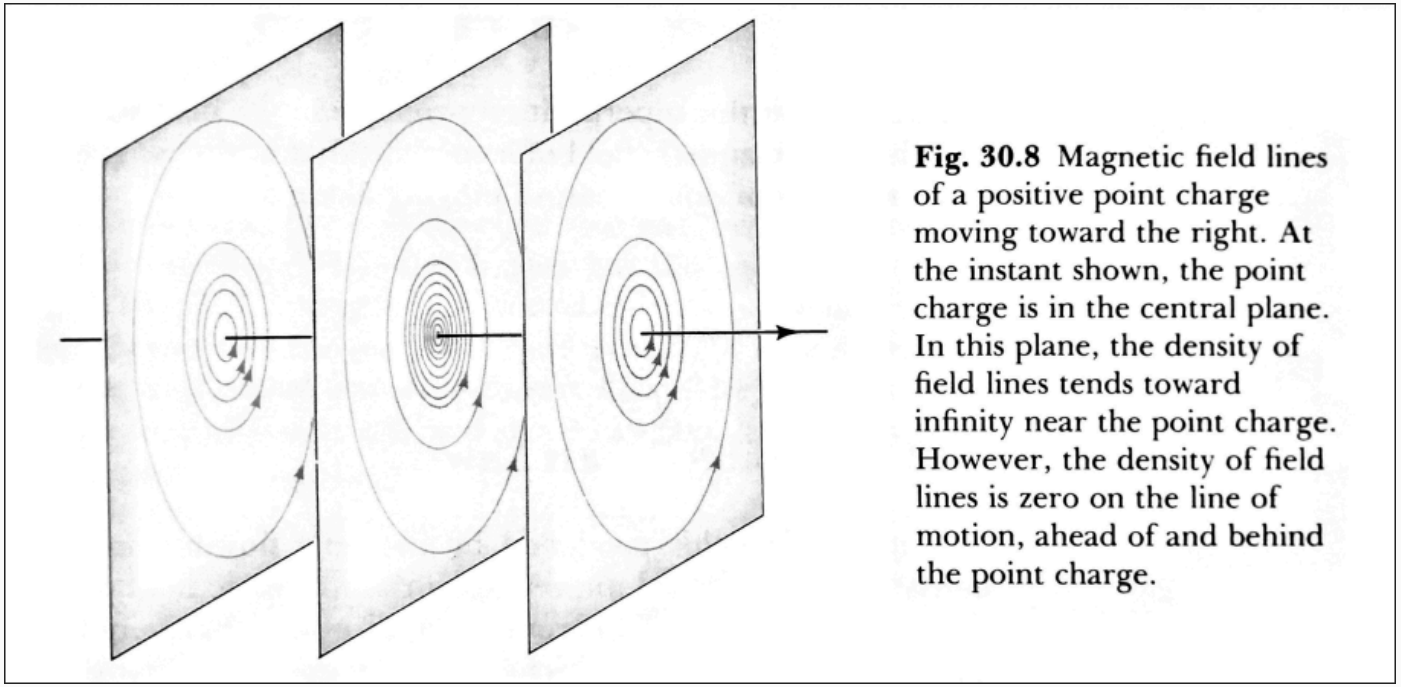


Fig. 30.8 Magnetic field lines of a positive point charge moving toward the right. At the instant shown, the point charge is in the central plane. In this plane, the density of field lines tends toward infinity near the point charge. However, the density of field lines is zero on the line of motion, ahead of and behind the point charge.

Magnetic field lines do not begin or end on charges as electric field lines do.

Gauss's law for magnetic fields

Magnetic field lines have no beginning or end. The net magnetic flux through a closed surface is zero:

$$\oint \vec{B} \cdot d\vec{S} = 0$$

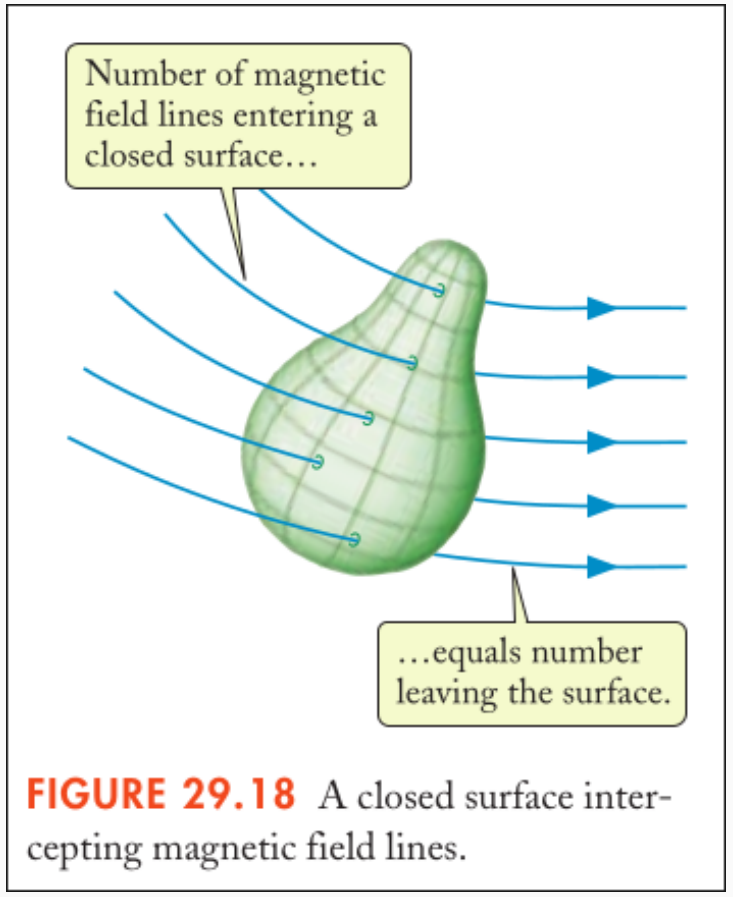


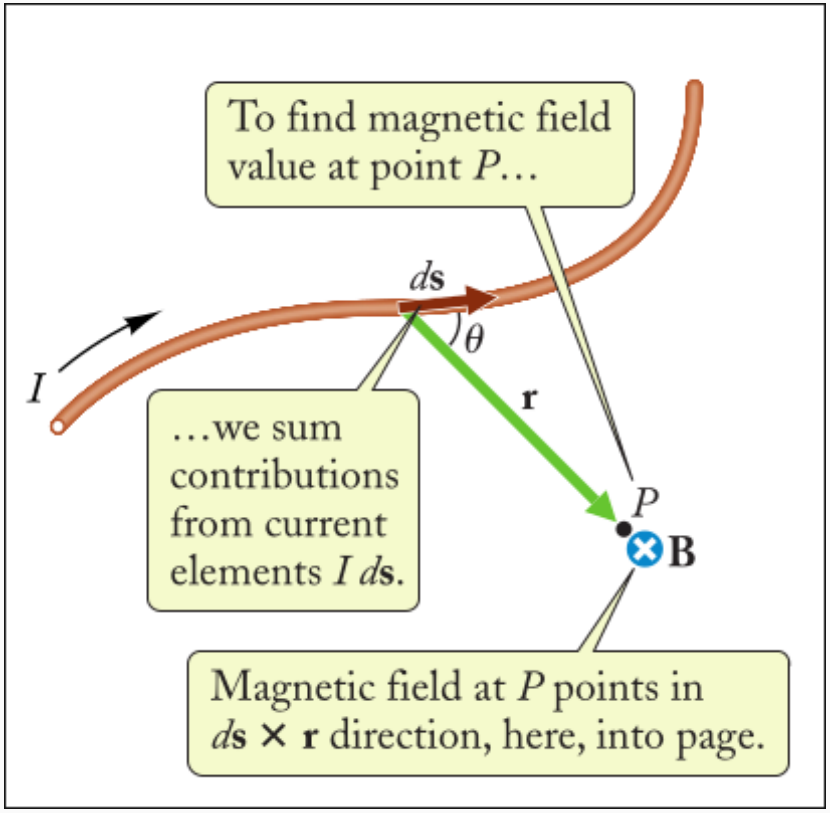
FIGURE 29.18 A closed surface intercepting magnetic field lines.

The Biot–Savart law

Fields produced by currents

A current in a wire is a distribution of moving charges, each generating its own magnetic field. By superposition, we can calculate the total field.

Start with a thin wire of arbitrary shape carrying a steady current. Assume the current is carried by positive charges, and consider a small section.



From moving charge to current

The small section contributes

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{dq'}{r^2}(\vec{v}' \times \hat{r})$$

Its magnitude is $dB = \frac{\mu_0}{4\pi}\frac{dq'}{r^2}v' \sin\theta$.

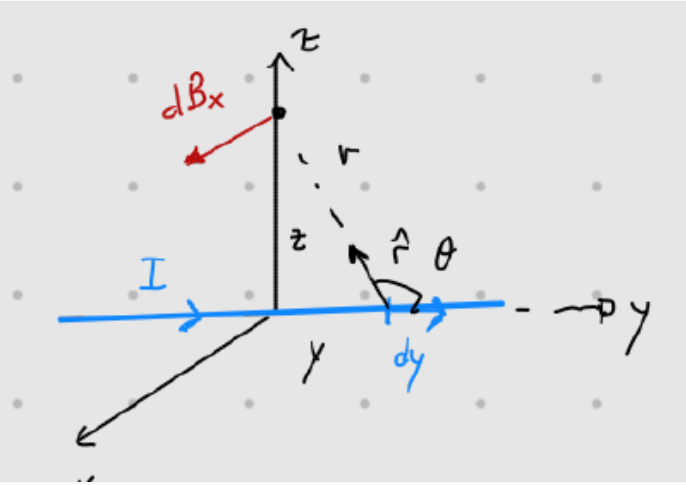
Use $dq' = I\,dt$ and $dt = \frac{dl}{v'}$:

$$dq' = \frac{I\,dl}{v'}$$

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l} \times \hat{r}}{r^2}$$

Example: a long straight wire

A very long, thin straight wire carries a steady current I. What is the magnetic field at a distance from its axis?



For a point at height z above the wire,

$$\mathrm{d}B_x = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}y\sin\theta}{r^2}$$

All contributions have the same direction:

$$B_x = \frac{\mu_0 I}{4\pi}\int_{-\infty}^{\infty}\frac{\sin\theta}{r^2}\,\mathrm{d}y$$

Long straight wire: evaluating the integral

Write everything in terms of θ:

$$y = z\cot(\pi - \theta) = -z\cot\theta$$

$$\mathrm{d}y = z\csc^2\theta\,\mathrm{d}\theta, \quad r = \frac{z}{\sin\theta}$$

The limits are $\theta = 0$ and $\theta = \pi$. Hence

$$B_x = \frac{\mu_0 I}{4\pi z}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta$$

$$\boxed{B_x = \frac{\mu_0 I}{2\pi z}}$$

Field lines around a straight wire

Fig. 30.11 (left) Magnetic field lines around a straight wire carrying a current. The wire is perpendicular to the plane of the page, and the current emerges from this plane.

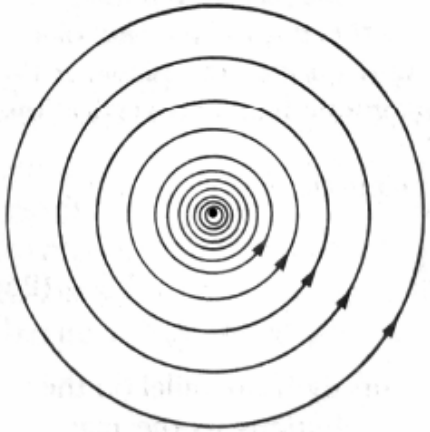
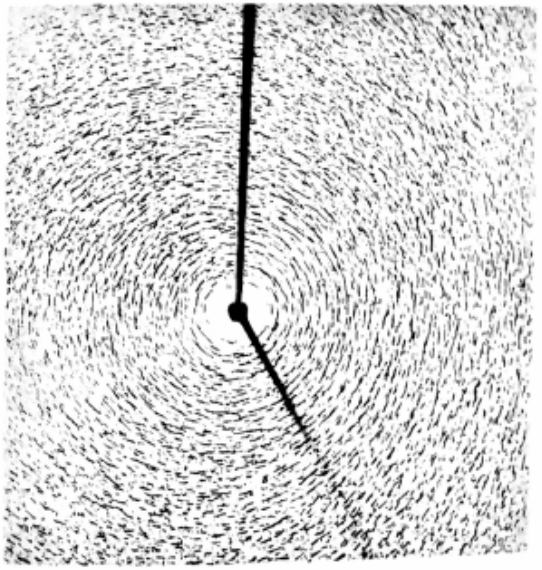
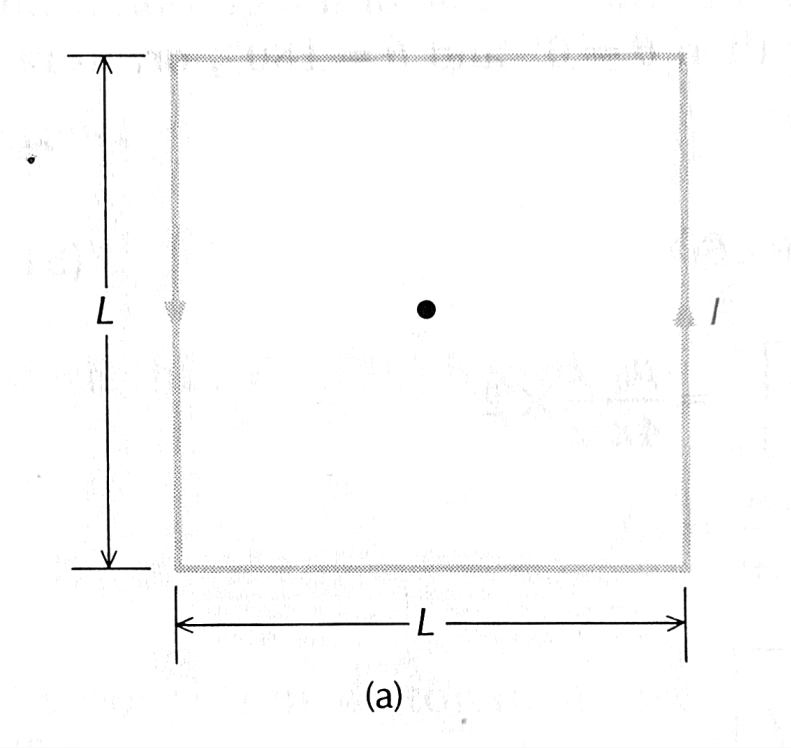


Fig. 30.12 (right) Magnetic field lines around a straight wire, made visible by iron filings sprinkled on a sheet of paper.



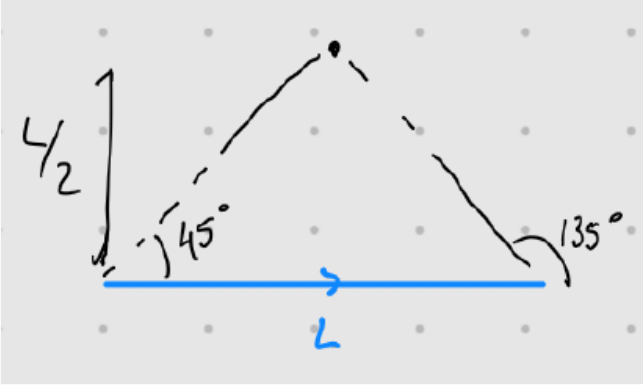
Example: a square loop

A square loop of dimensions $L \times L$ carries current I. What is the magnetic field at its centre?



The calculation is the same as for the infinite wire, but the angular limits are different for each side.

Square loop: the contribution of one side



$$B_1 = \frac{\mu_0}{4\pi}\frac{I}{\frac{L}{2}}\int_{\frac{\pi}{4}}^{3\frac{\pi}{4}} \sin\theta \, d\theta = \frac{\mu_0}{4\pi}\frac{I}{\frac{L}{2}}\sqrt{2}$$

Multiply by four to get the total field:

$$B = 4B_1 = \frac{\mu_0}{4\pi}\frac{8\sqrt{2}I}{L}$$

Example: field on the axis of a circular loop

Find the magnetic field on the axis of a circular loop of radius R carrying current I.

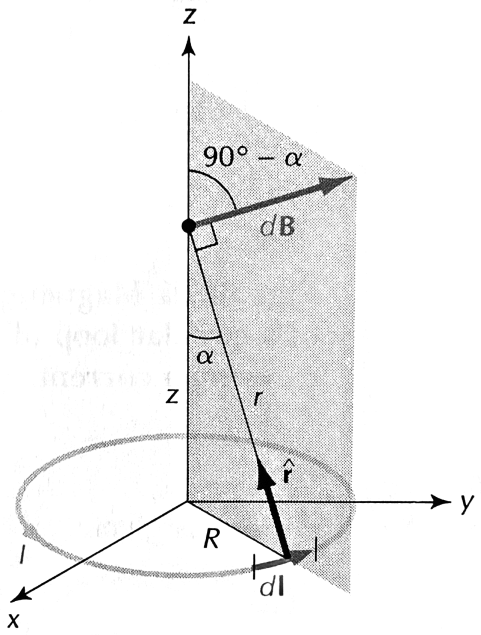


Fig. 30.14 Circular wire loop carrying a current I in the x–y plane.

$$dB = \frac{\mu_0}{4\pi}\frac{I\,dl}{z^2 + R^2}$$

Horizontal components cancel. Only the vertical component remains:

$$dB_z = dB\sin\alpha$$

$$\sin\alpha = \frac{R}{\sqrt{z^2 + R^2}}$$

Circular loop: integrating around the loop

$$B_z = \frac{\mu_0}{4\pi} \int \frac{IR\,\mathrm{d}l}{(z^2 + R^2)^{\frac{3}{2}}}$$

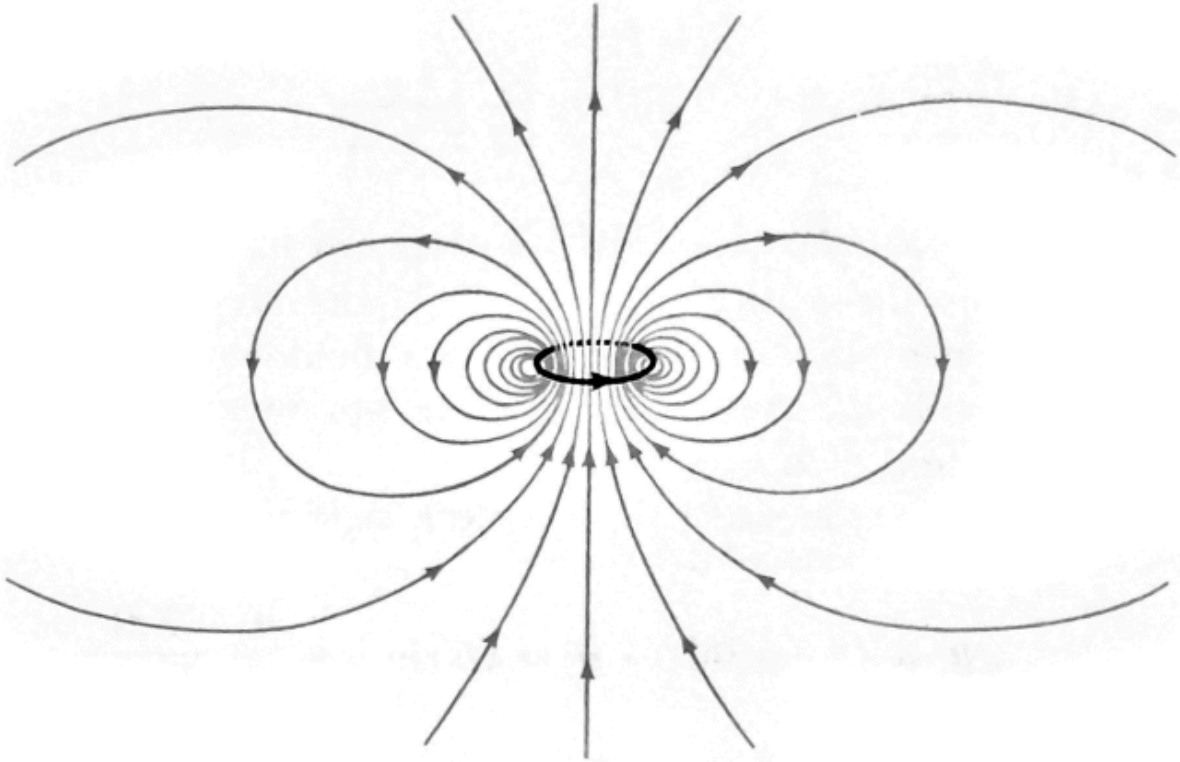
$$B_z = \frac{\mu_0}{4\pi} \frac{IR}{(z^2 + R^2)^{\frac{3}{2}}} \times 2\pi R$$

$$\boxed{B_z = \frac{\mu_0 I R^2}{2(z^2 + R^2)^{\frac{3}{2}}}}$$

This result is for points on the z-axis only.

Magnetic field lines of a circular loop

Fig. 30.15 Magnetic field lines of a circular loop of wire carrying a current.



Visualising the field with iron filings

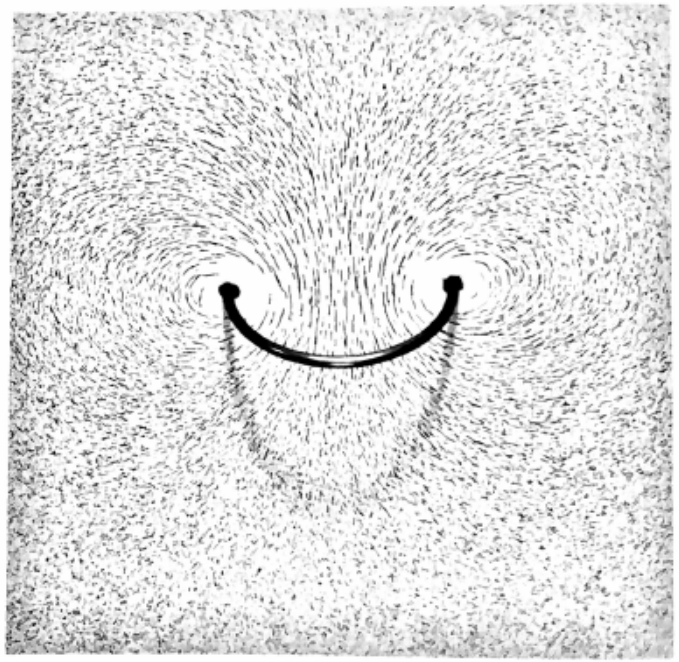


Fig. 30.16 Iron filings sprinkled on a sheet of a paper placed around a loop carrying a strong current.

The magnetic dipole

The magnetic dipole

At large distances, the magnetic field pattern of a current loop is similar to the electric field pattern of a dipole. A small current loop is therefore called a magnetic dipole.

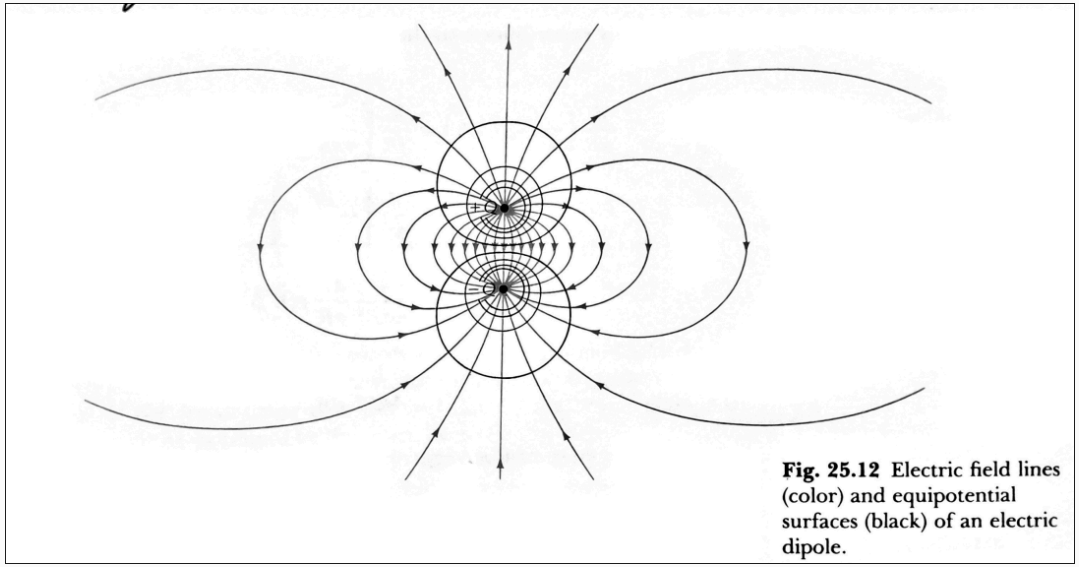


Fig. 25.12 Electric field lines (color) and equipotential surfaces (black) of an electric dipole.

Here small means that the loop size is small compared with the observation distance: $R \ll z$.

The magnetic dipole moment

Using the result for a circular loop,

$$B_z = \frac{\mu_0 I R^2}{2z^3\left(1 + \frac{R^2}{z^2}\right)^{\frac{3}{2}}}$$

For $R \ll z$,

$$B_z \approx \frac{\mu_0 I R^2}{2z^3}$$

It is customary to write $\mu = I\pi R^2$. Thus

$$\boxed{B_z \approx \frac{\mu_0}{2\pi}\frac{\mu}{z^3}}$$

μ is called the magnetic dipole moment of the loop.

Current times area

Note that μ has the form

$$\mu = IA$$

This is a general formula: it also gives the magnetic dipole moment of a rectangular current loop.

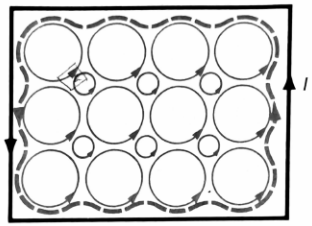


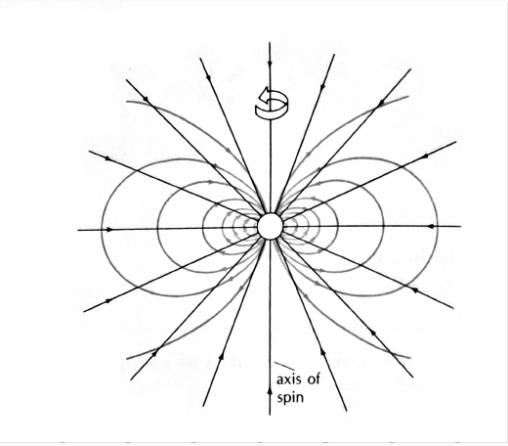
Fig. 30.17 A rectangular loop is approximated by an array of small circular loops. Each circular loop carries the same current as the rectangular loop. The dashed line represents the effective net current of the array of circular loops.

An arbitrary loop can be approximated by many small circular loops. Where adjacent loops touch, opposite currents cancel. Only the edge currents remain, reproducing the current around the outer loop.

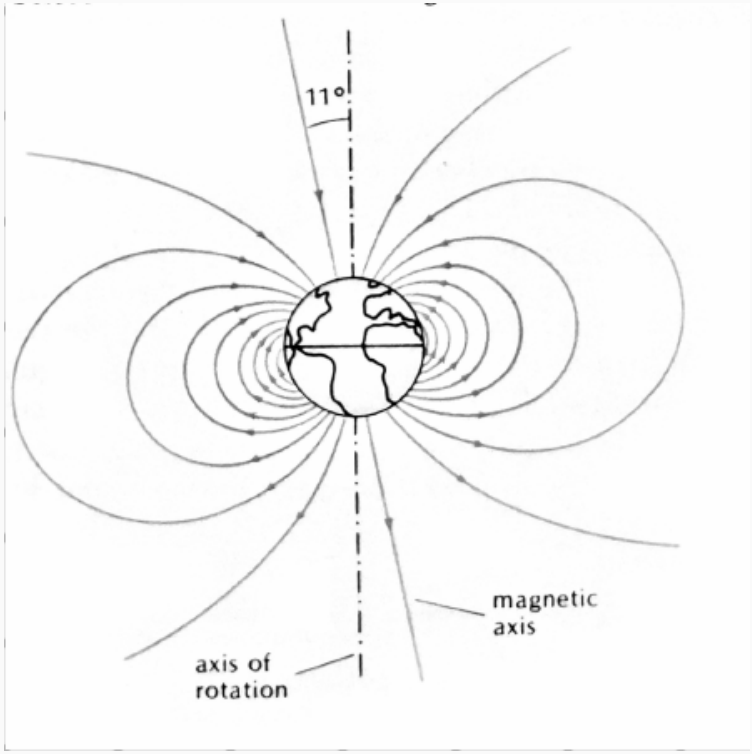
Magnetic moments of particles

Electrons, protons, and many elementary particles have magnetic dipole moments. Crudely, they may be pictured as small balls of charge spinning about an axis.

In reality, intrinsic particle magnetic moments are a quantum-mechanical effect.

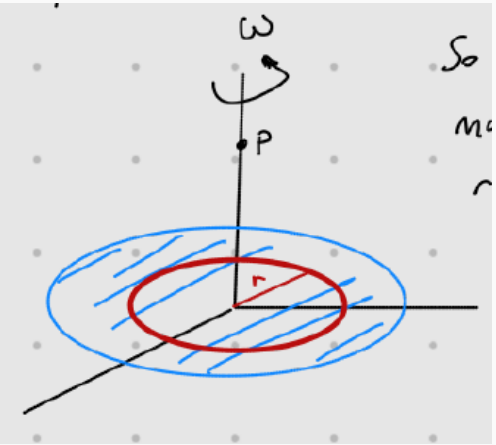


Earth’s magnetic field



Example: magnetic field of a spinning disk

A charge Q is uniformly distributed over a disk of radius R. The disk spins about its axis with angular velocity ω. Find the magnetic field at a point on its axis.



Think of the disk as loops of radius r, from 0 to R.

Each loop contributes

$$\mathrm{d}B_z = \frac{\mu_0 r^2}{2(z^2 + r^2)^{\frac{3}{2}}}\,\mathrm{d}I$$

Spinning disk: current in each ring

The surface charge density is $\sigma = \frac{Q}{\pi R^2}$. For a ring of width $\mathrm{d}r$,

$$\mathrm{d}q = \sigma 2\pi r \, \mathrm{d}r$$

The current is charge times frequency:

$$\mathrm{d}I = \mathrm{d}q \frac{\omega}{2\pi} = \frac{Q\omega r \, \mathrm{d}r}{\pi R^2}$$

$$B_z = \frac{\mu_0 Q \omega}{2\pi R^2} \int_0^R \frac{r^3 \, \mathrm{d}r}{(z^2 + r^2)^{\frac{3}{2}}}$$

Spinning disk: evaluating the integral

Integration by parts can work. Note that

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(-\frac{r^2}{\sqrt{z^2+r^2}}\right) = -\frac{2r}{\sqrt{z^2+r^2}} + \frac{r^3}{(z^2+r^2)^{\frac{3}{2}}}$$

Therefore

$$\int_0^R \frac{r^3\,\mathrm{d}r}{(z^2+r^2)^{\frac{3}{2}}} = -\frac{R^2}{\sqrt{z^2+R^2}} + \int_0^R \frac{2r\,\mathrm{d}r}{\sqrt{z^2+r^2}}$$

Use $u = z^2 + r^2$, $\mathrm{d}u = 2r\,\mathrm{d}r$.

Spinning disk: the axial field

For $z \geq 0$, the integral becomes

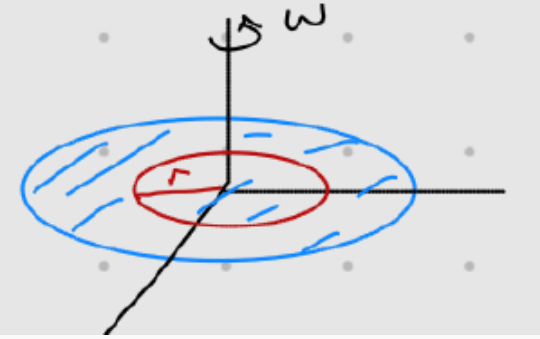
$$-\frac{R^2}{\sqrt{z^2+R^2}} + 2\sqrt{z^2+R^2} - 2z$$

Finally,

$$B_z = \frac{\mu_0 Q \omega}{2\pi R^2}\left(\frac{2z^2+R^2}{\sqrt{z^2+R^2}} - 2z\right)$$

Example: dipole moment of a spinning disk

A charge Q is uniformly distributed over a disk of radius R, spinning with angular velocity ω. Find its magnetic dipole moment.



Think of the disk as a collection of rings. The charge on a ring is

$$\mathrm{d}q = \frac{Q}{\pi R^2} 2\pi r \,\mathrm{d}r = \frac{2Qr\,\mathrm{d}r}{R^2}$$

Spinning disk: integrating the dipole moment

The current in a ring is

$$\mathrm{d}I = \frac{2Qr\,\mathrm{d}r}{R^2}\frac{\omega}{2\pi} = \frac{Qr\omega\,\mathrm{d}r}{\pi R^2}$$

Its magnetic moment is

$$\mathrm{d}\mu = A\,\mathrm{d}I = \pi r^2\frac{Qr\omega\,\mathrm{d}r}{\pi R^2} = \frac{Q\omega}{R^2}r^3\,\mathrm{d}r$$

Integrating over the disk,

$$\mu = \int_0^R \frac{Q\omega}{R^2}r^3\,\mathrm{d}r = \frac{Q\omega R^2}{4}$$